

6(a)	e.m.f.: energy transferred from chemical to electrical (per unit charge)	B1
	p.d.: energy transferred from electrical to thermal (per unit charge)	B1

Question	Answer	Marks
6(b)(i)	1 $I = 4.8 / 32$ $= 0.15 \text{ A}$	A1
	2 $P = EI$ or $P = VI$ or $P = I^2 R$ or $P = V^2 / R$ $= 6.0 \times 0.15$ or $0.15^2 \times 40$ or $6.0^2 / 40$	C1
	$= 0.90 \text{ W}$	A1
	3 number $= It / e$ $= [0.15 \times 25] / 1.6 \times 10^{-19}$	C1
	$= 2.3 \times 10^{19}$	A1
	or $Q = 0.15 \times 25 (= 3.75)$ number $= 3.75 / 1.6 \times 10^{-19}$	(C1)
	$= 2.3 \times 10^{19}$	(A1)
	4 $4.8 / 6.0 = 32 / (R_{XY} + 32)$ or $1.2 / 6.0 = R_{XY} / (R_{XY} + 32)$ or $4.8 / 1.2 = 32 / R_{XY}$	C1
	$R_{XY} = 8.0 \Omega$	A1
	Alternative methods:	
	$R_{XY} = (6.0 - 4.8) / 0.15$ or	(C1)
	$= 8.0 \Omega$	(A1)
	or $6.0 = 0.15 (32 + R_{XY})$	(C1)
	$R_{XY} = 40 - 32$ $= 8.0 \Omega$	(A1)

Question	Answer	Marks
6(b)(i)	5 $1 / 8.0 = 1 / R_X + 1 / 24$	C1
	$R_X = 12 \Omega$	A1
	Alternative method:	
	$I_Z = 4.8 / 32 = 0.15$ and $I_Y = 1.2 / 24 = 0.05$ $I_X = 0.15 - 0.05 (= 0.10)$	(C1)
	$R_X = 1.2 / 0.10 = 12 \Omega$	(A1)
6(b)(ii)	<u>total</u> resistance decreases	M1
	(so voltmeter) reading increases	A1

Question	Answer	Marks
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